[Lecture Note] Analyzing Quantitative Data

GE2021 Research Technology, Spring 2025

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Learning Objectives

- 10-1 Summarize why statistics are used in research methods
- 10-2 Conduct univariate analysis
- 10-3 Explain the measures of central tendency
- 10-4 Define measures of variability and dispersion
- 10-5 Describe the various ways to graphically represent data
- 10-6 Conduct bivariate analysis

SHORT VIDEO INTRODUCTION

https://www.youtube.com/watch?v=nHpf5V1SpNc

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https://www.youtube.com/watch?v=EUeQRE5UJpg

Analyzing Quantitative Data

- Statistics: a branch of mathematics that uses mathematical logic to interpret phenomena and draw conclusions from the accumulated data.
- It establishes relationships between and among variables and can even attempt to predict future behaviors or relationships.

Univariate Analysis

- Univariate analysis: the analysis of only one variable.
- Descriptive analysis: allows researchers to sketch the details of variables and gain familiarity with the sample; gives an idea of the descriptive characteristics of variables.
  - Descriptive statistics describe the first impressions of a variable.
- Inferential analysis/statistics: provide a deeper understanding of the variables because it draws conclusions about the population from the sample at hand.
  - This type of analysis goes beyond the description of the data available and attempts to draw relationships or associations between variables.
  - Once a relationship is detected between variables, inferential statistics attempts to predict the possibility of applying the same relationship to the entire population of the study.

- Inferential statistics operate by trying to calculate the random error that occurs during collection and analysis.
- Random error: associated with the type of common mistakes or errors that happen by chance when data are collected or entered into a spreadsheet or during analysis.
- Sampling error: a random error that occurs during the process of data collection.
  - Inferential statistics only investigates the sampling error that can happen randomly; it is unable to identify a systematic sampling error.

Univariate analysis is the first step researchers take to familiarize themselves with the population sample.

- Frequency Distributions
  - Constructing a frequency distribution with data is a simple procedure that can illustrate key characteristics of participants in the sample.
  - The first indicator is absolute frequency, which reflects the number of times an event or instance has been repeated.
  - Assigning percentages to each category adds another layer of valuable information.
    - Percentages are capable of zooming out from the raw numbers because it compares the raw numbers to the entire sampling population.
  - Cumulative percent: another descriptive measure of frequency; shows the collective percentage of categories together.
    - It is sensitive to the order of organization.

Measure of Central Tendency

Measures of central tendency: summarize an entire variable or category in a single value that is numerically representative of it.

- Mean: the mathematical average of a set of numbers; all numbers in a variable category are added together and then divided by n, which is the total number of entries in that data set.
  - The mean is useful for comparing two or more groups.
  - The mean takes into account every single point in the variable, and it is very sensitive to outliers.
  - Outlier: a data point that is much higher or lower than the majority of other points.
- Median: represents the center point in a set of numbers that have been arranged in ascending or descending order.
  - The median is useful in statistics because it gives an understanding about the distribution of a variable.
- Mode: the most frequently used number in a data set; which number is most commonly encountered in a specific set of numbers.
  - The mode can provide additional information that can potentially clarify any misleading information from the mean or median.
  - One variable can have more than one mode.
    - Unimodal: only one mode
    - Bimodal: two modes
    - Multimodal: three or more modes

Measures of Variability and Dispersion

- Measures of central tendency tell about the middle scores in a distribution, but they tell very little about how they are spread out.
- Measures of variability and dispersion are the ones that give information about how far apart or close by the values are in a distribution.
- Variability expresses the degree to which data in a data set spreads out, or deviates, from the center.

- Homogenous: sets that have little variability
- Heterogenous: sets with a lot of variability
  - Pose a higher level of complexity in approaching analysis.

- Range: the difference between the largest and smallest numbers in the set.
  - Interquartile range (IQR) is another measure of the middle 50% of the scores in a variable
- Variance: represents each datapoint's average deviation from the mean; the average sum of the squared distance of every single point in the data set from the mean.
- Standard deviation: the square root of the variance; also represents each data point's average deviation from the mean.

Normal Distribution

- The normal distribution is a theoretical perfect distribution where the mean, mode, and median are equal to one another.
  - It is symmetrical; the mean divides the distribution into equal halves.
  - It is used to compare against the actual distribution.
- Standard scores: also called z scores; indicate the number of standard deviations in a distribution and what proportion of the scores fall within this range.
- The proportion of z scores that fall between z scores in any normal distribution, regardless of its means or standard deviations, are defined by the following rules:
  - The area between z scores 0 and -1 and between 0 and 1 always equal 0.3413.
  - The area between z scores 1 and 2 and between -1 and -2 always equal 0.136.
  - The area between z scores 2 and 3 and between -2 and -3 always equal .0215.
  - The proportion of scores greater than z score 3 or smaller than z score -3 is 0.0013.
- These rules mean that approximately 68% of the data points fall between one standard deviation below and above the means.
  - 99% fall between 2 standard deviations below and above the mean.

Skewness and Kurtosis

- Skewness: when a distribution lacks the symmetry created when the mean, median, and mode are in the center; its shape is asymmetric when viewed in a histogram.
  - The term used to note that a distribution is clustered on the right or the left.
  - A distribution with a long tail on the left side and a majority of data on the right is called left-skew, left-tailed, or negatively skewed distribution.
  - A distribution with a long tail on the right but a majority of points of the left is called a right-skew, right-tailed, or positively skewed distribution.
  - In right/positively skewed distributions, the mode will be on the left of the distribution where the highest peak is, the median will have a greater value than the mode, and the mean will have a higher value than the median and mode.
  - The opposite occurs in left/negatively skewed distributions; the mean will have the smallest value, followed by the median, and then the mode, which will be at the highest peak.

- Kurtosis: refers to the flatness of the distribution and the flatness of its tails.
  - The two characteristics work together; a high peak in the distribution also has long flat tails.
    - Leptokurtic: distribution with a high peak and long flat tails
    - Platykurtic: distribution with a flat peak and short tails.
  - Kurtosis is an indication that some problematic outliers are present in the distribution.
  - If kurtosis is noted during analysis, the researcher will often take another look at the distribution to possibly remove outliers and bring the distribution to a shape closer to normal distribution.

Continuous and Discrete Variables

- Discrete variables: categorical variables that can only take whole numbers and do not have any continuity in terms of measurement.
  - Example: number of employees in an institution
- Continuous variable: a type of numerical variable that can take all possible units
  - Examples: income, height, weight

V. Graphical Representation of Data

A graphic representation of information can have a powerful impact on a broader audience, visually depicting relationships and patterns among variables.

- Bar graphs: use vertical bars that represent each variable in a study.
  - The viewer can easily distinguish not only which category occurs most frequently, but how it compares to the other categories.
- Histograms: often used for quantitative variables where the categories are related to one another.
  - Presents the frequency of a variable in the form of distinct but connected bars.
- Line graph: a line connecting points in a distribution.
  - Useful in depicting two or more categories simultaneously, or if there are multiple distributions in the same category.
  - A line graph is an effective way to demonstrate relationships between variables.
- Pie charts: a visual representation of the distribution of one variable.
  - Work best with categorical variables that have a small number of categories.
  - The main advantage of a pie chart is its visual impact on the audience.

VI. Bivariate Analysis

- Bivariate analysis: used to investigate the relationship between two variables; particularly, to understand whether the changes in one variable are reflected by changes in another.
- Correlation: used to determine the extent to which two quantitative variables approximate a linear relationship.
- Regression: used to identify the line that best describes this relationship.
- The Linear Model
    - i. A linear relationship between variables is a perfect relationship between two variables that, when graphed, resembles a straight line.
    - ii. In a linear relationship, the changes in units of one variable are reflected in specific interval changes in the other variable.
    - iii. They are rare in research findings because most of the variables collected from participants will have variability in the answers.

Correlation

- When all the data are cleaned and entered in a software program, scientists will typically run an analysis using a Pearson correlation coefficient, called r—an index that determines the level of correlation between two variables
- Pearson correlation coefficient r can range in value from −1 to 1, indicating the level of linearity between two variables.
    - The closer this coefficient, r, is to −1 or 1, the more likely it is that the relationship resembles a linear relationship.
    - The closer this coefficient is to 0, the less likely it is that these two variables are correlated to each other.
    - Correlations of 1 or −1 are perfect positive and negative relationships, respectively, while a correlation coefficient of 0 indicates no relationship between variables.

Causation

- The fact that two variables are strongly correlated with one another does not mean that one variable causes changes in the other variable.
- A spurious relationship happens when the correlation between two variables is affected by a third variable that interferes with the relationship we are studying.

Regression

- Regression/best fitting line: represents the closest possible linear relationship between two variables.
- Regression is used to predict future behaviors of the variables as much as to explain current behaviors.
  - The best fitting line of the relationship between variables also tells us the direction of a relationship

Scatterplots

- Scatterplots visually depict a bivariate relationship.
- Scatterplots are dots on a graph that represent a relationship between two quantitative variables: dependent and independent.
  - Each single dot represents one case.
  - The dependent variable is on the vertical axis (or y-axis) and the independent variable on the horizontal axis (or x-axis).

Class Activities:

- In this activity, you will be gathering and organizing data collected from students in your class. Use the table below to collect this information (ask students to handwrite their answers into one table that you circulate among all the students) and add additional rows to accommodate your class size. Photocopy the completed table and circulate it to all your students. Ask each student to enter the data exactly as collected from the table into a program like Excel or Stata for analysis.
  - Student name What is your height in inches?
  - What is your age in years?
  - What is your eye color? What is your hair color?
  - What is your favorite ice cream flavor?
  - Where were you born?

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Using the data gathered from the exercise above, ask students to generate a presentation of the results in PowerPoint or a similar program. Have a discussion as a class about the best way to present the data for each variable. What measures will you use to summarize the findings for each variable? How will you present the data (e.g., in graphical format).

- A researcher is working with a community partner to collect quantitative data on the effectiveness of a cardiovascular disease risk intervention for men of color. She is invited to present at a community meeting where participants do not understand the value of quantitative data collection and are more interested in data in the form of "stories." In a small group, discuss the benefits of quantitative data analysis and discuss what information you will share with the participants to convince them of the value of this type of analysis.

House Keeping

Announcement

Assignment

Assignment1: Assignment Title

- **What to do**
  - A
  - B
- **Format and Requirement**
  - PDF format, < ?? pages
  - file name should be include your student id and name
    - stuID_name_title.pdf (e.g. 1111111_ChungilChae_SelfIntroduction.pdf)
  - APA, Double-Space, No title page, 12 points, Reference section if necessary
  - Chinese name in English, English name, Student ID, Class name and section (e.g. GE2021, W09; MGS3001, W01)

- due date
  - by Date(DAY) 11:59PM
  - NO LATE SUBMISSION ALLOWED!!!!

Reference